

FITARNA

Evaluation 6

Presentation

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Questions?

Milestone 6

Milestone 6

Task Matrix

Task	Completion	Dathan	Ethan	Jacob	Vincenzo
1. Finalize Tour Mode	100%	0%	20%	40%	40%
2. Create a QR code for the .apk download	100%	0%	0%	50%	50%
3. Finalize Poster	100%	20%	20%	40%	20%
4. Conduct Eval and Analyze Results	100%	40%	20%	20%	20%
5. User Manual	100%	25%	25%	25%	25%
6. Demo Video	100%	0%	50%	50%	0%

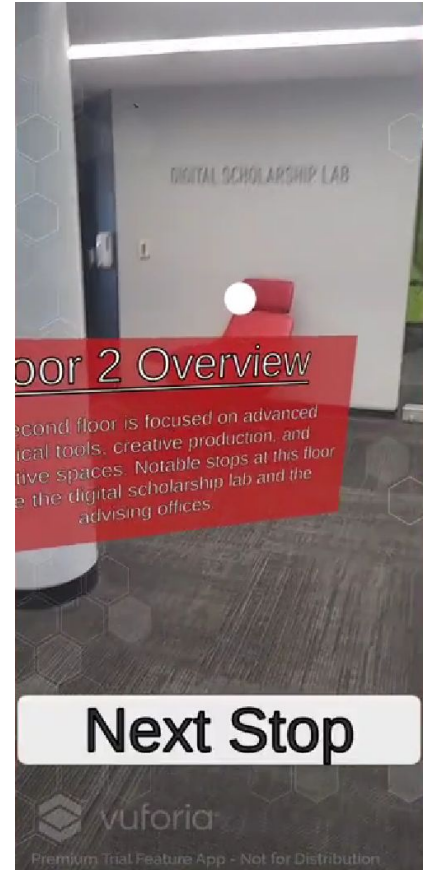
Task 1: Finalize Tour Mode

During Milestone 6, the team finalized Tour Mode for FITARNA.

This included:

- Integrating guided navigation waypoints throughout the FIT library
- Adding and organizing the tour content for each stop
- Ensuring smooth transitions between stops
- Verifying that AR overlays correctly displayed location labels and directional cues.

The feature was tested across multiple devices to confirm stability before the final release.



Task 2: Create a QR code for the .apk download

- Created a QR code for installing an Android APK
- Tested Installation for an old instance of FITARNA



Task 3: Finalize Poster

- Reviewed multiple poster layout ideas as a team
- Selected and completed the final design
- Organized the board into two columns
- Included Motivation, Project Goals, Implementation/Approach, and Software Used



FIT AR Navigation App (FITARNA)

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Motivation

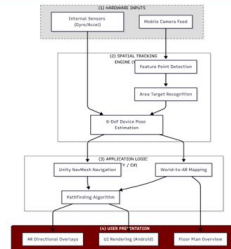
- Navigation Complexity: Large-scale academic facilities like the Evans Library present significant wayfinding challenges for new students and visitors.
- GPS Limitations: Standard satellite-based navigation fails indoors due to signal attenuation and lack of floor-level granularity.
- The Solution: FITARNA leverages Augmented Reality to provide intuitive, real-time visual guidance, bridging the gap between digital maps and the physical environment.



Implementation and Approach

- Environment Mapping: Conducted extensive spatial scans of the Evans Library using Vuforia Area Targets to create a high-fidelity digital twin for persistent tracking.
- Hybrid AR Framework: Integrated Unity's AR Foundation with the Vuforia Engine to leverage cross-platform ARCore/ARKit features while maintaining superior object and area recognition.
- Pathfinding Logic: Implemented a custom NavMesh within Unity to calculate real-time, shortest-path routes between navigational waypoints.
- Spatial UI Overlay: Developed a user interface that projects 3D directional markers and "breadcrumbs" onto the physical world, ensuring intuitive wayfinding in a complex indoor environment.

System Architecture



Project Goals

- High-Precision Localization: Achieve sub-meter indoor positioning accuracy without the use of GPS by leveraging AR spatial mapping.
- Intuitive AR Wayfinding: Develop a seamless navigation experience using real-time 3D directional overlays.
- Dynamic Point-of-Interest System: Implement a searchable database of campus locations, such as specific library wings, study rooms, and help desks.

Software Stack



Task 4: Conduct Evaluation and Analyze Results

- Performed user evaluation of the FITARNA app
- Collected feedback on usability, navigation accuracy, and overall experience
- Analyzed responses to identify remaining pain points
- Results showed users were able to navigate the library successfully using the app

Task 5: User Manual

- Collaboratively wrote both a user manual and developer manual
- User manual covers installation, launching the app, Tour Mode usage, and troubleshooting
- Developer manual explains system architecture, AR framework integration, build steps, and future maintenance guidance

Task 6: Demo Video

- Demo video created by Ethan and Jacob
- Showcased full FITARNA functionality on a physical Android device



Questions?